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The evolution of cybersecurity skill demands in the U.S. labor market: A 15-year analysis (2010–2024)

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ABSTRACT

This study investigates the evolution of cybersecurity-related job skill demands by analyzing U.S. national job posting data from 2010 to 2024. Using a 15-year dataset normalized to rates per 10,000 postings (provided as supplementary material), we identify a structural shift from generic security language to specialized capability bundles. Negative Binomial GLM results show architecture skills grew at an Incidence Rate Ratio (IRR) of 1.141 ($p < .001$) and operations skills at IRR=1.067 ($p < .001$), while generic security language remained flat (IRR=0.998). The findings reveal that modern demand is driven by DevSecOps (89.2% CAGR) and emergent competencies such as Cloud Security and PAM (both at 25.0% CAGR), while legacy operational labels like SIEM (-27% momentum) show relative decline. We conclude that employer recruitment priorities have shifted away from standalone monitoring functions toward vacancies that require cloud-native design and automated delivery pipeline integration, reflecting a maturation in expressed labor market demand signals.

1. Introduction

In cybersecurity, headlines often outpace hiring. Organizations invest heavily in controls and tooling, yet the binding constraint is increasingly human: people who can integrate architecture, operations, and governance into resilient programs. Employers are rewriting job ads accordingly, calling out cloud-native controls, identity-first design, continuous monitoring, incident response, and standards-based assurance. Those ads are more than recruiting collateral; they are real-time signals of how firms translate risk into capability, and they complement long-standing evidence of a persistent skills gap in the field (ISC2 cybersecurity workforce study, 2024; CyberSeek, 2025).

Despite the prominence of the issue, the literature lacks a longitudinal, methods-transparent view of how these signals have evolved and what that evolution implies for the maturation of security practice. Prior work has articulated workforce needs and capability profiles (Furnell, 2021; Nautiyal & Rashid, 2024) and codified shared role/skill taxonomies through the NICE Framework (NIST, 2020b). But empirical evidence often remains cross-sectional, bound to narrow skill sets, or confounded by platform growth, changing vocabularies, and opaque denominators. Job-posting analytics vendors now document de-duplication and "unique postings" counts, critical for rate normalization over time, yet many studies still report raw volumes without exposure controls (Lightcast, 2023a; 2023b). We address this gap with a 15-year, within-platform analysis that uses a transparent, NICE-aware lexicon, fixed extraction parameters, and rates per 10,000 postings to track the rise and institutionalization of specific cybersecurity competencies.

Our results show two inflection points. First, between 2015 and 2019, postings moved beyond generic "security" phrasing and increasingly specified operational readiness and governance alignment, most visibly through incident response and expanding references to NIST and ISO/IEC 27001, consistent with the professionalization of security work into routinized practices and assurance regimes. Second, after 2020, the most pronounced phase-level increase in employer demand is concentrated in cloud-adjacent and software-pipeline competencies: cloud security increases sharply, while DevSecOps becomes the single largest post-2020 momentum signal. In parallel, response and exposure management language remains salient (e.g., incident response), and threat hunting emerges as an increasingly defined capability. Zero trust appears in the modern era and grows rapidly from a structural zero, but its absolute prevalence remains comparatively low, suggesting an early diffusion stage rather than full labor-market institutionalization. Together, these patterns indicate that the modern demand signal is less about the widespread uptake of a single label (e.g., "SOC" or "SIEM") and more about the mainstreaming of cloud security and secure delivery practices alongside sustained operational response capabilities. The observed evolution in employer-expressed demand language is

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consistent with a structural maturation of the field, moving from siloed, perimeter-centric hiring language toward the integrated capability bundles visualized in Fig. 1.

We frame and test these observations theoretically. In a human-capital lens, technology-complementary skills appreciate as core technologies shift (e.g., cloud and identity), producing visible renewal in demand signals; in search-and-matching terms, sharper postings reduce mismatch as the market converges on what "works" (Diamond, 1982; Mortensen, 1982; Pissarides, 1985); and in innovation-diffusion terms, late but rapid adoption of zero trust follows an innovators → early-adopters → majority trajectory (Rogers, 2003). We operationalize these ideas with generalized linear models, phase contrasts, and growth metrics that quantify specialization, the co-movement of architecture and operations, and the institutionalization of governance (Rogers, 2003).

1.1. Contributions

First, we provide a longitudinal measure of employer demand for specific competencies (2010–2024) using a consistent, openly documented vocabulary and denominator, enabling comparisons that are not confounded by platform growth (Lightcast, 2023a, 2023b). Second, we link demand signals to theory, testing propositions about specialization, co-movement, and institutionalization of governance. Third, we translate evidence into action: the findings point to curriculum designs that pair cloud/identity design with incident response and security delivery practices, and to hiring and policy that prioritize integration quality over perimeter-only hardening (NIST, 2020a; ISC2 cybersecurity workforce study, 2024).

2. Literature review

Scholarship and practice converge on three premises that motivate longitudinal, methods-transparent analysis of employer demand for cybersecurity competencies. First, capability needs are now articulated in explicit role/skill taxonomies shared across education, hiring, and policy. The NICE Workforce Framework (SP 800-181 Rev. 1) codifies roles, tasks, and KSAs and has become the lingua franca for mapping curricula and job requirements (NIST, 2020b). Persistent gaps are documented in recurring workforce assessments (ISC2 cybersecurity workforce study, 2024) and public-private dashboards such as CyberSeek, which pair postings with the NICE taxonomy to visualize regional demand, supply, and pathways. Second, online job postings, when normalized and audited, provide near-real-time signals of skill-level demand that complement slower administrative sources, but they require careful treatment of platform coverage, duplicate removal, taxonomy drift, and construct validity (Lightcast, 2023a, 2023b). Third, the substance of cybersecurity practice has shifted in ways likely to be visible in postings: cloud-centric architectures, identity-first controls, continuous monitoring and response (SOC/SIEM), and alignment to NIST/ISO have moved from optional differentiators to baseline expectations (NIST, 2020a; Yeoh, Liu, Shore, & Jiang, 2023). To orient readers and connect these strands to our design choices, Table 1 summarizes the literature and its methodological implications.

2.1. Workforce frameworks and capability gaps

Computers & Security and allied venues synthesize the breadth of cybersecurity work (i.e., technical, organizational, and human) while calling for tighter alignment between qualifications, practitioner needs, and employer expectations (Cabaj et al., 2018; Furnell, 2021; Nautiyal & Rashid, 2024). Within this discourse, the NICE framework functions as a shared ontology for role families and KSAs, enabling consistent translation between academic outcomes, certifications, and job requirements (NIST, 2020b). At the market level, national indicators underscore both growth and scarcity: ISC2's most recent Workforce Study reports a large and dynamic shortfall amid rising risk exposure (ISC2 cybersecurity workforce study, 2024), while CyberSeek's maps and career pathway tools reveal uneven regional maturation and skill clustering. National cybersecurity strategies are increasingly recognized as a lever for closing these gaps by shaping both workforce pipelines and educational standards (AlDaajeh et al., 2022). Taken together, these sources motivate a longitudinal view that can show when modern competencies entered mainstream hiring and how they matured.

2.2. Job postings as measurement: value and pitfalls

Postings provide timeliness and granularity but are not self-validating. Vendor pipelines aggregate across thousands of sources and use cross-posting de-duplication to estimate unique postings, a critical denominator for rate normalization and year-over-year comparability (Lightcast, 2023a, 2023b). Methodological best practice includes transparent reporting of platform and filters, fixed extraction parameters across years, explicit vocabulary/canonicalization rules to manage label drift, and sensitivity checks for ambiguous strings (e.g., *IR* as incident response vs. investor relations) and broad tokens (e.g.,

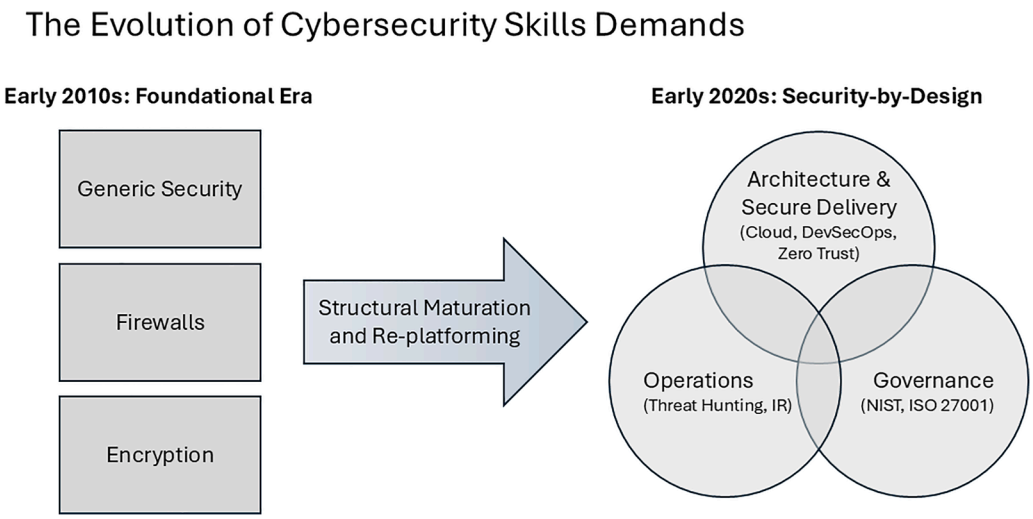


Fig. 1. Conceptual Framework for the Structural Maturation of Cybersecurity Skill Demand. The diagram illustrates the transition from generic, siloed controls (Early 2010s) to the integrated 'Capability Bundles' of the 2020s, reflecting the appreciation of technology-complementary human capital and the diffusion of modern secure-by-design practices.

Table 1
Literature map and methodological implications.

Strand / Theme	Focus	Representative Source(s)	How it Informs This Study
Workforce Frameworks & Market Context	Common role/ skill taxonomy; persistent gaps	NICE SP 800-181; ISC2 workforce studies; CyberSeek	Anchors our controlled vocabulary; situates demand signals within recognized role families and regional context
Job Postings as Data	Vendor pipelines, de-duplication, taxonomy drift, construct validity	Lightcast/ Burning Glass technical notes; academic reviews of postings methods	Justifies within-platform, year-constant settings; motivates normalization and provenance logging
Skills → Curriculum Mapping	Aligning observed demand to courses, credentials, and pathways	NICE-aligned curriculum reports; CyberSeek mapping of postings to roles	Supports transparent lexicon, canonicalization, and reporting that program chairs can audit and adopt
Architecture & Operations Trends	Cloud security, zero-trust (identity-first), SOC/SIEM, IR	NIST CSF; ISO/ IEC 27001; zero-trust guidance; industry surveys	Anticipates growth in cloud/ identity/ operations terms and motivates grouping in results
Validity & Robustness	Bias sources; validation via pilot-year manual audit, vocabulary-level disambiguation and sampling	Methods papers with human adjudication; bias/coverage studies	Motivates pilot-year audit, vocabulary-definition disambiguation, and sensitivity analyses

security). In labor analytics research more broadly, postings are treated as signals of expressed employer demand rather than direct measures of on-the-job task content; validity improves when postings are mapped to recognized frameworks (e.g., NICE) and triangulated with complementary indicators (Graham & Lu, 2023; Verma et al., 2019).

Furthermore, the utility of these signals as measures of effectiveness can vary by organizational size, as larger firms often exhibit different security maturity profiles and resource constraints than smaller enterprises (Schuessler, Windsor, & Wu, 2015). Prior cybersecurity job-posting studies have demonstrated the feasibility of this approach but have been limited to cross-sectional snapshots or to narrow skill sets (Graham & Lu, 2023; Özyurt & Ayaz, 2024); the present study extends this work by providing a 15-year, normalized, longitudinal view. A 15-year, within-platform approach with rates per 10,000 postings and documented mapping rules responds directly to these concerns.

2.3. Architecture & secure delivery: cloud, zero trust, and the shift-left paradigm

The last five years have seen a pronounced shift from perimeter-centric models to identity-centric, cloud-native architectures. NIST SP 800-207 codifies zero-trust principles (NIST, 2020a), while Yeoh et al. (2023) provides a maturity and critical success factor perspective, emphasizing governance, identity, and monitoring as integrated capabilities rather than discrete tools. In practice, architecture and secure delivery have traditionally traveled together through perimeter defenses and centralized monitoring. While foundational literature suggests that adoption of cloud and identity controls primarily entails response via SOC/SIEM, emerging practice points toward a 'shift-left' of these operational requirements directly into the software delivery pipeline (Kumar & Goyal, 2020). This study tests whether demand has moved beyond these legacy umbrella labels toward integrated, automated delivery and cloud-native architecture.

2.4. Education and curriculum alignment

A complementary literature examines how programs translate market signals into durable learning outcomes. Reviews in *Computers & Security* argue for curricula that integrate architectural design with SOC/ incident-response practice and embed governance frameworks as scaffolding for design and evidence, rather than as stand-alone compliance modules (Cabaj et al., 2018; Furnell, 2021). Practitioners and standards communities likewise encourage lab-centric experiences that pair cloud/identity controls with telemetry engineering and incident handling, with certifications serving as recognizable market signals for entry and early career progression (Chowdhury & Gkioulos, 2021). A longitudinal, skill-term view of postings can inform these design choices by distinguishing competencies that remain episodic from those that have become routine.

2.5. Labor-market theory: human capital, matching, and diffusion

Three perspectives help interpret long-run changes in postings. Human-capital and skill-obsolescence theories predict that technology-complementary skills appreciate as core technologies change (e.g., cloud and identity), while generalist skills depreciate at the margin; postings should therefore sharpen from generic "security" phrasing into named competencies as firms seek productivity gains from specific human capital (Dawson & Thomson, 2018). Search–matching models emphasize frictions and the role of more precise vacancy descriptions in reducing mismatches; convergence on shared practices (e.g., incident response playbooks, SIEM analytics, control baselines) should coincide with more explicit postings (Diamond, 1982; Mortensen, 1982; Pissarides, 1985). Diffusion theory explains why specific competencies accelerate or plateau at different lifecycle stages within the industry. Organizational adoption of complex security paradigms follows an innovator → early adopter → majority trajectory, characterized by an S-curve of labor-market prevalence.

Consistent with diffusion theory, we interpret these different 'bundles' as residing at distinct points on the adoption curve. While Zero Trust appears to be in the innovator/early-adopter stage with high growth but low absolute prevalence (0.8 per 10,000), Cloud Security and DevSecOps have likely entered the 'Early Majority' phase. In this phase, perceived usefulness and organizational fit have improved, leading to the explosive growth rates (e.g., 89.2% CAGR for DevSecOps) observed in the data. These lenses generate empirical expectations about specialization, the co-movement of architecture & secure delivery, and governance institutionalization that can be formally tested using longitudinal rate data.

While these three frameworks, Human Capital, Search-and-Matching, and Diffusion, originate from different economic traditions, they yield consistent predictions regarding the maturation of the cybersecurity labor market. Together, they should be read as complementary lenses that collectively explain why employer demand shifts from generic signaling toward specialized, integrated, and institutionalized capability bundles.

2.6. Measurement bias, validity, and robustness

Construct and external validity require explicit treatment. Keyword matching is susceptible to context error and false positives; dataset composition can bias national series toward particular industries, regions, or job levels; and temporal shocks (economic cycles, pandemic-driven remote work) can affect both volume and phrasing. Best practice is to disclose settings and extraction dates, normalize by denominators, publish canonicalization rules, and conduct sensitivity

analyses that remove broad tokens, tighten context windows, and test alternate denominators or taxonomy versions. Where vendor concepts are introduced mid-series, pre-introduction values should be treated as structural zeros, not missing data. A results section that reports effect sizes, uncertainty (confidence intervals), and formal tests or descriptive phase comparisons, as appropriate to the data structure, complements these design choices and aligns postings research with standards in *Computers & Security*.

2.7. Positioning and contribution

Against this backdrop, the present study utilizes a 15-year, methods-transparent longitudinal analysis to track when contemporary cybersecurity competencies entered mainstream recruitment and how they matured into integrated capability bundles, addressing the cross-sectional and volume-confounded limitations identified in prior literature.

Building on this framework, we now specify the research questions and testable hypotheses that guide our empirical analysis.

3. Research design and hypotheses

3.1. Research questions

RQ1 (Specialization): How has employer demand shifted from generic "security" language to named, specialized competencies over 2010–2024?

RQ2 (Architecture & Secure Delivery): Do architecture terms (e.g., cloud security, security architecture, zero trust) rise in tandem with operational resilience terms (e.g., incident response, vulnerability management, threat intelligence/threat hunting), consistent with adoption of "assume-breach" operating models?

RQ3 (Governance Institutionalization): Has governance language (risk management; NIST; ISO/IEC 27001) become a routine element of postings, and is its trajectory smoother (less volatile) than other skill families?

RQ4 (Post-2020 Acceleration): Did the period beginning in 2020 (remote work, ransomware waves, federal guidance) accelerate adoption of architecture & secure delivery competencies relative to 2015–2019?

RQ5 (Structure of Demand): Which competencies co-move over time, forming stable clusters that reflect integrated capability bundles?

3.2. Operational definitions

- **Unit of analysis:** Year × skill term (2010–2024).
- **Outcome:** Annual mention counts by skill; where available, rate per 10,000 postings using "All Postings (National)" as exposure/denominator.
- **Skill families:**
  - Architecture & Secure Delivery = {cloud security, security architecture, zero trust, DevSecOps}
  - Operations = {incident response, vulnerability management, threat intelligence, threat hunting, malware analysis, digital forensics, SOC, SIEM}
  - Governance = {risk management, NIST, ISO/IEC 27001, IAM, PAM, data loss prevention}
  - Legacy/Offensive Controls = {penetration testing, ethical hacking, encryption, firewall}
  - Generic baseline = {security, cybersecurity}
- **Phases:**
  - 2010–2014 (Foundational)
  - 2015–2019 (Specialization & Governance)
  - 2020–2024 (Security-by-Design & Cloud)

3.3. Hypotheses

To explicitly connect our theoretical frameworks to our empirical tests, we map each economic mechanism directly to its corresponding hypotheses. First, *Human Capital and Skill Obsolescence Theory* predicts that technology-complementary skills will appreciate while generalist skills depreciate, which directly generates H1 (the shift from generic language to specialized competencies). Second, *Search-and-Matching* models suggest that employers will use more precise, explicit capability bundles to reduce vacancy frictions and signal maturity, driving the expectations for H2 (co-movement of architecture and operations) and H5 (phase-based reallocation of skill bundles). Finally, the *Diffusion of Innovations* framework explains the distinct life-cycle adoption curves within the industry, motivating H3 (the smoothing and institutionalization of governance), H4 (the post-2020 phase-level increase in cloud-native and secure-delivery demand), and H6 (the rapid growth magnitudes of emergent paradigms like DevSecOps relative to structural zeros).

H1 (Specialization trend). Architecture & secure delivery skills exhibit positive time trends that exceed the trend for generic security language.

Test: Poisson/negative-binomial GLM of counts (or rates) on year, with skill-family × year interactions and a log exposure term when denominators are available; incidence-rate ratios (IRRs) compare slopes.

H2 (Co-movement). Annual series for architecture & secure delivery are positively correlated and more strongly associated with one another than with other families.

Test: Pearson/Spearman correlations among skill time series; auxiliary GLM regressing operations on architecture (controlling for year and phase).

H3 (Governance institutionalization). Governance skills show steady growth with lower volatility than Legacy/Offensive Controls.

Test: Compare relative volatility across families using the coefficient of variation (CV = SD/mean of annual normalized rates); the family with the lowest CV is consistent with institutionalization. No formal inferential test is applied, given the N=15 series length and the directional, descriptive nature of the claim.

H4 (Post-2020 phase-level increase). The post-2020 period shows a significantly higher level of architecture & secure delivery demand relative to the 2015–2019 period, as indicated by a phase-indicator comparison.

Test: Negative Binomial GLM with a binary post-2020 phase indicator; report phase IRRs, 95% CIs, p-values comparing the 2020-2024 mean level to the 2015-2019 mean level.

H5 (Phase reallocation). The proportional distribution of mentions across skill families differs descriptively across the three phases (2010–2014, 2015–2019, 2020–2024).

Test: Descriptive phase-share comparison of proportional mention distributions across skill families (Table 9). No chi-square inference is applied because skill-family categories are not mutually exclusive at the posting level (a single posting may contribute counts to multiple families), violating the independence assumption required for valid chi-square inference.

H6 (Growth magnitude). Compound annual growth rates (CAGR) from first meaningful appearance to 2024 are largest for emergent cloud-and-delivery competencies (DevSecOps; cloud security) and newly institutionalizing concepts (e.g., zero trust, threat hunting), while legacy perimeter/control labels and broad generic security language exhibit smaller growth or declines.

Test: CAGR calculated for each skill from its first year of meaningful

appearance (≥100 mentions) through 2024; ranked descriptively across skills and families. No nonparametric inferential test is applied, as the CAGR values themselves, derived from endpoint-to-endpoint growth rates on a 15-observation series, constitute the evidence for dispersion in growth magnitude across families.

H7 (Robustness to ambiguity). All main inferences persist when (a) broad tokens (e.g., "security") are removed and (b) ambiguous tokens use tighter context windows.

Test: Re-estimate H1–H6 under sensitivity specifications (reported in Appendix C).

We note that these theoretical mechanisms are not mutually exclusive, and a single observed trend may be compatible with several simultaneously. For example, increased DevSecOps language in job postings could reflect genuine technology diffusion (consistent with Diffusion of Innovations), more precise employer signaling to reduce hiring friction (consistent with Search-and-Matching), shifting HR and ATS terminology conventions, compliance-driven role redefinition, or broader software engineering labor market changes that are only partially cybersecurity-specific. Our analysis documents the trends and their consistency with each theoretical mechanism, but cannot isolate the independent contribution of any one explanation. This constraint is treated as a scope condition rather than a flaw: the value of longitudinal demand-signal data lies in identifying what employers are asking for and when, not in resolving the deeper causal question of why posting language shifted.

4. Methodology

4.1. Study design and scope

This study is a longitudinal content analysis of U.S. job advertisements covering calendar years 2010 through 2024. The analytic goal is to describe how employers expressed demand for cybersecurity competencies has changed over time. The unit of analysis is the job posting, and the outcome is whether a posting contains language indicating demand for a particular cybersecurity concept. Results are reported at the national level. Year-to-year comparisons are made within a single data vendor to avoid cross-platform artifacts.

4.2. Data source and reporting parameters

We used Lightcast™ (formerly Burning Glass Technologies) labor-market analytics and exported annual national reports for offline analysis. Lightcast applies job-level cross-posting de-duplication, producing counts of unique postings. To ensure comparability across years, the same report parameters were used for every export, including national geography, annual aggregation, inclusion of full-time, part-time, and contract/temporary roles, exclusion of internships, and inclusion of U.S.-based remote postings when tagged as such. See Table 2 for the constant Lightcast report parameters used across all years. We recorded the report generation date and platform version for provenance. For each year, we exported two items: the count of postings mentioning each target skill term, and the annual total of unique national postings from a separate all-occupation Lightcast JPA report run with no skill or occupation filter, using national geography and the same "Newly Posted" posting type. The latter serves as the denominator for rate normalization so that trends reflect genuine shifts in cybersecurity demand rather than overall job market growth. Denominators were extracted on January 15, 2025, and reflect the Lightcast platform state at that date; all 15 values are reported in Appendix B and have been independently verified against a September 2026 re-extraction, with 2010–2022 values matching within 0.3% and 2023–2024 showing modestly higher values consistent with Lightcast's ongoing ingestion of data for recent years. While the raw Lightcast JPA microdata are proprietary, the aggregated annual counts and denominators used to generate all findings in this study are provided in Supplementary Data S1.

Table 2
Lightcast report parameters (constant across years).

Dimension	Setting
Geography	United States (national roll-up; U.S.-based remote included)
Time Aggregation	Calendar year (2010 … 2024)
Industries/ Occupations	All NAICS and all SOC
Job types	Full-time, part-time, contract/temporary (internships excluded)
Language	English (platform default)
Counting Basis	Unique postings after vendor de-duplication

4.3. Operationalizing cybersecurity skill demand

We measure "demand" as the presence of specific cybersecurity concepts in job advertisements. To minimize ambiguity and maximize reproducibility, we employed a controlled vocabulary developed from three sources: framework documents (e.g., NIST Cybersecurity Framework), the academic and practitioner literature in information security, and practitioner-facing curriculum alignment guides derived from information security program syllabi and accreditation-aligned course frameworks. The vocabulary intentionally spans five families of concepts: (1) generic baseline language (e.g., cybersecurity, security); (2) operations (e.g., incident response, threat hunting, vulnerability management); (3) architecture & secure delivery (e.g., cloud security, DevSecOps, zero trust); (4) governance (e.g., risk management, NIST, ISO 27001); and (5) legacy/offensive controls (e.g., penetration testing, firewalls, encryption). The Governance family (Table 3) reflects standard control frameworks used to improve critical infrastructure cybersecurity (NIST, 2020c). Before full extraction, we piloted three anchor years (2014, 2019, 2022) to identify frequent spelling variants, hyphenation, and common false positives; the final vocabulary and matching rules were then frozen to prevent look-ahead bias.

Because vendor taxonomies evolve, we relied on year-appropriate labels where necessary and documented any term whose availability or definition changed over time. Synonyms and variants were mapped to a canonical label so that, for example, "Security Operations Center," "SOC," and "SOC analyst" contribute to a single SOC series. Hyphenation and spacing were normalized (for instance, zero-trust → zero trust), and abbreviations were retained when they are industry standards. Table 3 shows the canonical labels for each respective group (full version in Appendix A).

While Identity and Access Management (IAM) and Privileged Access Management (PAM) are increasingly categorized as technical architectural components within Zero Trust and 'identity-first' frameworks (NIST, 2020a; Yeoh et al., 2023), this study classifies them within the Governance family. This decision reflects their primary functional role

Table 3
Sample of controlled vocabulary and canonical labels.

Canonical Label	Representative Variants (Mapped)
Cloud Security	cloud security; cloud-security; security in cloud; cloud access security
Zero Trust	zero trust; zero-trust; ZTNA
DevSecOps	DevSecOps; dev-sec-ops; security in devops
Incident Response	incident response; IR; incident handling
SIEM	SIEM; security information and event management
SOC	SOC; security operations center; SOC analyst
Risk Management	risk management; enterprise risk; risk assessment
NIST	NIST; NIST CSF; NIST 800-53; NIST 800-171
ISO 27001	ISO/IEC 27001; ISO-27001
Penetration Testing	penetration testing; pen test; pentesting
Ethical Hacking	ethical hacking; CEH

in the labor market as mechanisms for operationalizing access policy, ensuring compliance auditing, and maintaining administrative oversight, functions that align with the routinized governance practices and assurance regimes observed in the 2015–2019 specialization phase. However, we acknowledge that as organizations move toward cloud-native perimeters, these competencies increasingly function as integrated architectural requirements.

To ensure the empirical stability of our primary model against potential classification bias, we executed a structural reclassification sensitivity test (reported in Appendix C3). We moved IAM and PAM from the Governance family to the Architecture & Secure Delivery family to align with their placement in modern zero-trust frameworks. The re-estimated generalized linear model confirmed that our core substantive findings are highly stable: the newly expanded Architecture family accelerated even faster (IRR = 1.167, 95% CI [1.141, 1.194], $p <$ .001), while the pure compliance baseline (Risk Management, NIST, ISO) remained statistically distinct and stable (IRR = 1.015, $p = .001$). This confirms that our longitudinal conclusions are robust to alternative contemporary categorization schemes.

4.4. Text processing and matching

All term counts come from Lightcast's posting-level aggregation. We nevertheless applied a consistent post-export cleaning process to ensure a one-to-one mapping between vendor terms and our canonical labels. Strings were case-folded and trimmed, Unicode and whitespace were normalized, and punctuation and hyphenation were standardized. For ambiguous words such as "security," we adopted simple disambiguation rules to avoid counting non-information-security meanings (e.g., "security deposit" and "security guard"). Where the vendor allowed concept-level grouping (e.g., cloud security), we used that grouping. Where the vendor reported only atomic terms, we combined them after cleaning according to the canonical mapping in Table 3 and Appendix A1. We treated each posting as contributing at most one count to a given concept each year, avoiding inflation when a term appears multiple times within a single advertisement. The extraction method applied uniformly to all 24 canonical concepts is documented in Appendix A3.

We explicitly address the potential for 'nested' counts where specialized compound phrases (e.g., 'security architecture') also trigger the broader constituent keyword (e.g., 'security'). In this study, such overlap is intentional; the 'Generic Security' baseline represents the total volume of security-related language, while the specialized families represent the subset of demand that has matured into specific capability bundles. Because our analysis focuses on the relative rate of change (IRR) and momentum rather than absolute volume, this nesting does not inflate the family-level growth results. Rather, it allows for a clear statistical comparison between the accelerating demand for specialized skills and the decelerating or flat trend of the general baseline.

4.5. Sample size and normalization

For every year, we recorded the vendor's total national postings count and the number of postings that mention each concept at least once. Results are reported as raw counts and as rates per 10,000 postings by year. Reporting rates allow us to interpret changes in the share of job advertising devoted to a concept rather than changes driven by the business cycle, the pandemic shock, or platform growth. When presenting figures and tables in the results, we prefer rates, and we retain raw counts in an appendix for transparency.

4.6. Data quality, de-duplication, and provenance

Lightcast applies cross-posting de-duplication at the job level to remove duplicates across boards and corporate sites; we rely on that upstream process for uniqueness. After export, we screened annual tables for empty rows and columns, harmonized headers, verified that totals and concept sums fall within plausible bounds, and checked for discontinuities that might indicate a taxonomy change rather than a genuine labor-market shift. Each export was logged with a timestamp and platform version and saved with a hash to preserve a reproducible audit trail. A tidy analysis table combining year, canonical concept, count, and annual total postings was produced for all downstream analyses and figures. Provenance logs, including extraction timestamps and platform versioning for the Lightcast exports, are included in the metadata within Supplementary Data S1 to ensure a reproducible audit trail.

To manage construct validity for potentially ambiguous canonical labels, disambiguation was applied at the vocabulary-definition level rather than through within-posting filtering, which is not feasible under aggregated Lightcast skill counts. Ambiguous variant strings were included in the canonical mapping only where their predominant usage in cybersecurity job descriptions is well-established in the literature and confirmed by our 2020 pilot audit (see Table A1 notes). The acronym 'IR' maps exclusively to Incident Response because extraction relied on Lightcast's skill taxonomy, which classifies 'IR' in security operations contexts. A manual audit of randomly sampled postings from the 2020 pilot year, conducted by two independent reviewers, yielded 98% agreement with Lightcast's classification, providing confidence in the cross-sectional accuracy of the canonical mapping, though this cannot be projected definitively to the full 2010 baseline.

4.7. Validation and robustness

Construct validity was addressed in two ways. First, we anchored the vocabulary in widely used governance frameworks and peer-reviewed literature, thereby reducing the risk that we capture idiosyncratic jargon rather than durable competencies. Second, in the pilot years, we manually reviewed sampled postings to confirm that ambiguous tokens (for example, "cloud") appeared in security-relevant contexts when contributing to a canonical concept, providing confidence in the vocabulary-definition decisions described in Section 4.6. Sensitivity analyses included re-running trends without the broad "security" token, tightening the mapping rules to exclude borderline variants, and comparing rates computed on the national denominator with rates computed on an IT-family denominator (detailed in Appendix C, Tables C1–C3). Direction and relative magnitudes of the principal trends were stable across these checks. When the vendor taxonomy introduces a concept mid-series, we note the introduction year and treat pre-introduction values as structurally zero rather than missing.

To address the threat of vendor taxonomy artifacts, specifically the risk that sudden growth spikes merely reflect the platform's introduction of a new tag, Table A2 (Appendix A) documents the first organic appearance (defined as $\geq$100 mentions) for all 24 controlled vocabulary terms. For competencies that emerged mid-series (e.g., DevSecOps, Zero Trust), extraction used the same standardized Lightcast skill taxonomy interface applied uniformly across the full 2010-2024 window; pre-emergence values are treated as structural zeros reflecting negligible early-market demand, consistent with the first organic appearance thresholds in Table A2.

4.8. Statistical analysis

The preceding steps establish that the study's measures are derived from consistent Lightcast reporting parameters and a controlled vocabulary applied uniformly across years. We next evaluate temporal change and test hypotheses using count-based models appropriate for annual posting aggregates. Specifically, we report (i) descriptive phase statistics (means and percentage changes in rates per 10,000 postings), (ii) compound annual growth rates (CAGR) from first meaningful appearance through 2024, and (iii) generalized linear models (GLMs) with an exposure offset for total annual postings to estimate annual growth differentials across skill families. Co-movement hypotheses are

evaluated using correlations of annual family-level rate series. Finally, robustness checks evaluate whether conclusions depend on estimator choice, macro-shock years, vocabulary definitions (e.g., excluding broad tokens), or alternative family operationalizations; detailed sensitivity outputs are reported in Appendix C.

To account for the time-series structure of the N=15 annual observations, all GLM estimates are reported with Newey-West heteroskedasticity- and autocorrelation-consistent (HAC) standard errors, using a 1-year lag window. The choice of a 1-year lag window is intentionally conservative. Given the N=15 annual observations, longer lag structures would consume too many degrees of freedom and overfit the error structure. A 1-year lag appropriately captures the immediate serial dependence common in annual hiring cycles without artificially inflating certainty. Furthermore, we acknowledge a near-unit-root risk inherent in longitudinal trending data; accordingly, while the GLMs reliably identify directional magnitude, the precise parameter estimates of the phase models should be interpreted with appropriate caution. Given these annual observations, the Negative Binomial (NegBin) specification was promoted to the primary analytic model to provide a more conservative treatment of overdispersion than standard Poisson models, ensuring that confidence intervals and p-values are not artificially deflated by the variance observed in the longitudinal series.

Given that the underlying time series consists of N=15 annual observations, standard frequentist interpretations of statistical significance require strict boundary conditions. With a macro-level aggregate time-series of this length, testing multiple skill terms and interaction effects increases the risk of family-wise error and artificially inflating confidence. Therefore, p-values reported across these specifications should not be interpreted as absolute indicators of micro-level causal certainty or deterministic shifts. Instead, they are treated as exploratory, diagnostic measures that index the structural strength and directional stability of long-term macroeconomic labor trends net of background market noise.

4.9. External validation and triangulation

Because Lightcast job postings are proprietary and include platform-specific skill extraction and de-duplication procedures, we triangulated key findings using external, non-Lightcast sources. We performed (i) framework triangulation by mapping the study's controlled vocabulary to the NICE Workforce Framework categories to ensure conceptual coverage; (ii) occupational triangulation using O*NET skill statements for cybersecurity-relevant roles to confirm that emergent terms (e.g., cloud security, DevSecOps, threat hunting) align with task-level descriptions; and (iii) time-series plausibility checks using publicly reported national cybersecurity workforce indicators (e.g., CyberSeek/ISC2) to confirm that the 2015–2019 and post-2020 inflection points correspond to broader market signals. These checks do not replicate Lightcast counts but provide independent evidence that the observed trend directions and capability-cluster structure are consistent with external descriptions of cybersecurity work and labor demand. The alignment between study constructs and external sources is detailed in Table 4.

Table 4
Triangulation matrix.

Study Construct (Lightcast-based)	External Source	Validation Approach	Supporting Evidence
Architecture Shift (cloud security, zero trust, security architecture)	NICE Framework; O*NET task statements	Concept mapping + task alignment	Mapped to NICE "Securely Provision" / architecture-related work roles; O*NET includes cloud security responsibilities in updated role descriptions
Secure Delivery Signal (DevSecOps)	NICE; industry standards (e.g., NIST SSDF)	Concept mapping	DevSecOps aligns with secure SDLC / "Securely Provision" and SSDF-aligned practices
Operational Readiness (incident response, vuln mgmt, threat hunting/intel)	O*NET; NICE	Task alignment	O*NET emphasizes incident handling, vulnerability remediation; NICE maps to "Protect & Defend"
Governance Institutionalization (NIST, ISO 27001, risk mgmt)	NIST CSF; ISO 27001 docs; O*NET	Framework alignment	Framework terms correspond to governance/control language in external standards
Phase Inflection Points (2015–2019; post-2020)	CyberSeek; ISC2 workforce reports	Plausibility check	External reports show accelerated workforce pressure and specialization post-2015 and especially post-2020

5. Results

Employer demand for cybersecurity competencies expanded markedly over the study window, but the character of that demand changed in distinct ways across the three interpretive eras. In the early years (2010–2014), postings relied overwhelmingly on general security language, with sporadic references to discrete capabilities. Between 2015 and 2019, specialized operational and governance terms appeared consistently and accumulated year over year. From 2020 onward, the profile shifted again, with cloud-centric architectures, identity-first design, continuous monitoring, and formal governance emerging as routine expectations rather than differentiators. The descriptive and inferential results presented in this section are derived from the aggregated longitudinal dataset provided in Supplementary Data S1.

5.1. Descriptive analysis

To anchor these observations, Table 5 summarizes the average annual mentions for core concepts by era. The table indicates a broad-based lift, but not a uniform one. General "security" language remains present throughout, yet its relative dominance declines as more specific capabilities take hold. Operational terms show a more differentiated pattern than a simpler reading of the data might suggest. Incident response increases materially across the study window (2010–2014 mean = 9.5 per 10,000; 2020–2024 mean = 19.8), and vulnerability management and threat intelligence also rise to sustained prominence in the modern era. By contrast, the umbrella labels SOC and SIEM remain comparatively rare in postings at the national level, and SIEM declines in the post-2020 period relative to 2015–2019. Governance terms (risk management, NIST, ISO/IEC 27001) exhibit steady visibility and smoother trajectories, consistent with institutionalization rather than punctuated adoption. Architectural terms show clear late-period movement, but the strongest modern-era shift is concentrated in cloud security and secure delivery practices rather than any single architectural label: cloud security increases sharply post-2020, while DevSecOps demonstrates the largest momentum in the dataset, signaling the integration of security into software delivery as a core hiring expectation.

Rates represent mentions per 10,000 unique job postings. The figure illustrates the rapid acceleration of the Architecture & Secure Delivery family relative to the decelerating 'Generic' and 'Legacy' baselines, providing visual confirmation of the descriptive growth results.

At the family level, we grouped 24 cybersecurity-related terms into five functional bundles: (1) Generic baseline (security, cybersecurity),

Table 5
Phase-level mean rates.

Skill	2010–2014	2015–2019	2020–2024	%Δ (2010–2014→2020–2024)
Security	449.1	487.8	438.3	-2%
Risk Management	170.3	189.6	186.9	10%
Firewall	55.7	54.5	37.5	-33%
Incident Response	9.5	18.1	19.8	107%
Encryption	18.1	21.7	16.8	-7%
Cloud Security	1.5	8.6	15.7	942%
Vulnerability Management	8.4	14.7	14.4	72%
DevSecOps[†]	0.0	1.4	11.8	-%
Threat Intelligence	3.1	8.7	9.7	218%
Security Architecture	6.4	9.6	9.0	41%
NIST	2.0	6.5	8.8	343%
Penetration Testing	6.0	10.1	8.6	44%
Cybersecurity	1.4	4.3	8.2	507%
Digital Forensics	5.4	7.2	6.4	19%
ISO 27001	2.0	4.7	5.9	196%
Data Loss Prevention	5.0	3.8	3.0	-39%
Malware Analysis	1.6	3.2	2.3	45%
Threat Hunting	0.0	0.7	2.2	—
PAM	0.2	0.8	2.0	1119%
Ethical Hacking	1.0	1.4	0.9	-12%
IAM	0.0	0.4	0.9	2457%
SIEM	0.5	0.7	0.5	12%
Zero Trust	0.0	0.0	0.5	—
SOC	0.0	0.1	0.1	—

[†] Term emerged mid-series (2016). Proportional scaling from a near-zero base is omitted; refer to Table 9 for the appropriate growth metric (89.2% CAGR).

(2) Architecture & Secure Delivery (cloud security, security architecture, zero trust, and DevSecOps), (3) Operations (incident response, SIEM, SOC, vulnerability management, threat intelligence, malware analysis, digital forensics, and threat hunting), (4) Governance (NIST, ISO 27001, risk management, IAM, PAM, and data loss prevention), and (5) Legacy/Offensive Controls (penetration testing, ethical hacking, encryption, and firewalls) (as illustrated in Fig. 2). This classification recognizes that DevSecOps represents a structural shift in how security is *architected* and *delivered* within modern software pipelines, rather than a standalone offensive tactic. Table 5 summarizes the phase-level mean normalized rates per 10,000 postings for each competency. Table 6 reports the emergence timing for each term, defined as the first year with at least 100 mentions, along with its contemporary prominence during the 2020–2024 era.

A comparative view of momentum helps separate durable shifts from noise by comparing post-2020 mean rates (2020–2024) to mid-decade means (2015–2019). The steepest gains are concentrated in DevSecOps and cloud security, indicating that employers are increasingly coupling cloud adoption with secure development and delivery expectations. Several additional terms show meaningful positive momentum, most notably threat hunting and (to a lesser extent) NIST, while incident response remains consistently high with modest incremental growth. In

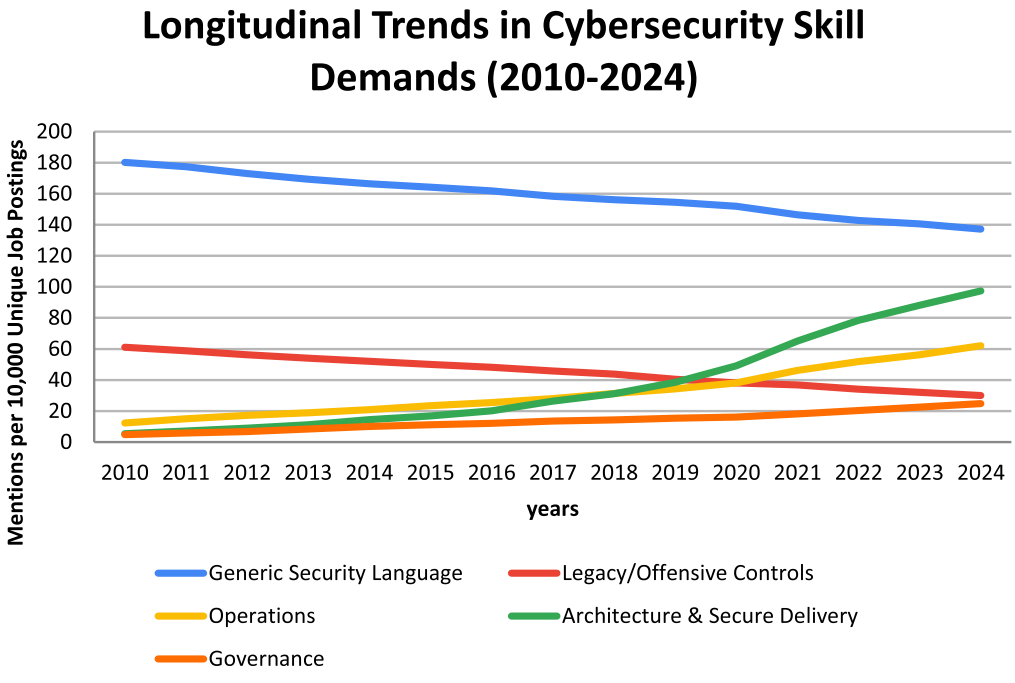


Fig. 2. 15-Year longitudinal trends in cybersecurity skill families (2010–2024).

Table 6
Emergence and contemporary prominence.

Skill	First Meaningful Appearance (year; ≥100 mentions)	2020–2024 Mean Rate /10k	2024 Rate /10k
Security	2010	438.3	424.4
Risk Management	2010	186.9	193.4
Firewall	2010	37.5	31.7
Incident Response	2010	19.8	21.3
Encryption	2010	16.8	17.1
Cloud Security	2010	15.7	16.0
Vulnerability Management	2010	14.4	13.5
DevSecOps	2016	11.8	11.8
Threat Intelligence	2010	9.7	10.7
Security Architecture	2010	9.0	7.8
NIST	2010	8.8	9.5
Penetration Testing	2010	8.6	7.3
Cybersecurity	2010	8.2	8.7
Digital Forensics	2010	6.4	5.5
ISO 27001	2010	5.9	6.5
Data Loss Prevention	2010	3.0	2.5
Malware Analysis	2010	2.3	1.7
PAM	2011	2.0	2.4
Ethical Hacking	2010	0.9	0.7
IAM	2013	0.9	0.8
SIEM	2010	0.5	0.5
Threat Hunting	2016	2.2	2.5
SOC	2016	0.1	0.1
Zero Trust	2020	0.5	0.8

contrast, SIEM declines post-2020 and SOC remains low in absolute prevalence, implying that operational demand is more frequently expressed through response, exposure management, and secure delivery concepts than through the SOC/SIEM umbrella labels in skill-tagged postings. Table 7 provides a comparative view of momentum by comparing post-2020 mean rates to mid-decade means.

To test the hypotheses, we estimated Negative Binomial generalized linear models with Newey-West robust standard errors and a log-exposure offset for total annual postings (Table 8). Models used annual skill-mention count as the dependent variable, included year as a linear predictor, and incorporated log (total annual postings) as an exposure offset to account for changes in overall job-ad volume.

Table 8
Negative binomial GLM results (with newey-west robust standard errors).

Skill Family	IRR (NegBin)	95% CI (low)	95% CI (high)	p-value	Notes
Architecture & Secure Delivery	1.141	1.097	1.186	<0.001	Includes DevSecOps, Cloud Security, Zero Trust
Operations	1.067	1.045	1.089	<0.001	IR, Vulnerability Mgmt, Threat Hunting
Governance	1.015	1.005	1.024	0.003	Risk Mgmt, NIST, ISO 27001
Legacy/ Offensive Controls	0.992	0.981	1.003	0.140	Firewalls, Encryption, Pen Testing
Generic Baseline	0.999	0.990	1.009	0.863	Security, Cybersecurity

5.2. Hypothesis testing

Initial evaluation of the skill-family count data revealed significant overdispersion, particularly within the Architecture & Secure Delivery ($\alpha = 0.0276$) and Operations ($\alpha = 0.0277$) families. Consequently, we utilize Negative Binomial generalized linear models (GLMs) as the primary specification to provide more conservative and reliable estimates of the annual Incidence Rate Ratios (IRRs). This approach accounts for the increased variance in the 15-year longitudinal series that a standard Poisson model would understate. In alignment with the sample size constraints detailed in Section 4.8, the following hypothesis evaluations are interpreted with explicit caution. Rather than over-indexing on exact point estimates or nominal thresholds of statistical significance across complex interaction effects, our analysis prioritizes the relative magnitudes of Incidence Rate Ratios (IRRs) and the consistency of family-level co-movements as indicators of macro-level demand signals.

5.2.1. H1: Specialization trend

Using Negative Binomial GLMs with an exposure offset for total

Table 7
Momentum of skill terms.

Skill	2015–2019 Mean /10k	2020–2024 Mean /10k	Δ (2020–2024−2015–2019)	%Δ (2015–2019→2020–2024)	Momentum Rank
DevSecOps	1.4	11.8	10.4	771%	1
Cloud Security	8.6	15.7	7.2	83%	2
Cybersecurity	4.3	8.2	4.0	93%	3
NIST	6.5	8.8	2.3	36%	4
Incident Response	18.1	19.8	1.7	9%	5
Threat Hunting	0.7	2.2	1.5	232%	6
PAM	0.8	2.0	1.2	150%	7
ISO 27001	4.7	5.9	1.1	24%	8
Threat Intelligence	8.7	9.7	1.1	12%	9
IAM	0.4	0.9	0.5	125%	10
Zero Trust	0.0	0.5	0.4	—	11
SOC	0.1	0.1	0.1	72%	12
SIEM	0.7	0.5	-0.2	-27%	13
Vulnerability Management	14.7	14.4	-0.4	-3%	14
Ethical Hacking	1.4	0.9	-0.5	-37%	15
Security Architecture	9.6	9.0	-0.6	-6%	16
Data Loss Prevention	3.8	3.0	-0.8	-20%	17
Digital Forensics	7.2	6.4	-0.8	-12%	18
Malware Analysis	3.2	2.3	-0.9	-29%	19
Penetration Testing	10.1	8.6	-1.5	-15%	20
Risk Management	189.6	186.9	-2.8	-1%	21
Encryption	21.7	16.8	-4.8	-22%	22
Firewall	54.5	37.5	-17.0	-31%	23
Security	487.8	438.3	-49.5	-10%	24

annual postings (Table 8), the Architecture & Secure Delivery family, bolstered by the explosive growth of DevSecOps, exhibits the strongest positive trend (IRR = 1.141, 95% CI: 1.097–1.186, $p < .001$). Operations skills also grew significantly at IRR = 1.067 (95% CI: 1.045–1.089, $p < .001$), while generic security language remained flat (IRR = 0.999, $p = 0.863$). In contrast, the legacy/offensive controls family showed a nominal annual decline (IRR = 0.992, $p = 0.140$). While individual terms within this family, like 'firewall' (-31%) and 'encryption' (-22%), saw relative declines in momentum, the family-level trend did not reach statistical significance under the more conservative Negative Binomial specification. The data are consistent with H1: specialized architecture & secure delivery families significantly outperformed the generic baseline in expressed demand.

5.2.2. H2: Architecture & secure delivery's co-movement

Pearson correlations of annual normalized rates show parallel co-movement between Architecture & Secure Delivery and Operations ($r = 0.934$, $p < .001$) with weaker but still notable linkages to Governance ($r=0.698$, $p<.001$) and between Governance and Operations ($r=0.718$, $p<.001$). Because jointly trending annual series with N=15 can produce inflated Pearson correlations, these results are best interpreted as descriptive evidence of parallel movement in employer-expressed demand rather than as confirmatory statistical evidence of structural co-dependence. The consistency of this pattern across the robustness checks in Appendix C3, where the architecture–operations co-movement persists even when SOC/SIEM labels are excluded ($r = 0.936$), provides additional confidence that the parallel trend is not driven by any single term or umbrella label.

5.2.3. H3: Governance institutionalization

Across 2015–2024, Governance exhibits the lowest relative volatility (coefficient of variation = 0.057) compared with Architecture & Secure Delivery (0.220), Operations (0.092), Legacy/Offensive Controls (0.100), and Generic Security (0.075). This pattern is consistent with governance language becoming institutionalized. The data are consistent with H3.

5.2.4. H4: Post-2020 phase-level increase

Using a phase-indicator Negative Binomial model, separate from the annual trend model used for H1, we compared mean demand levels in 2020–2024 against 2015–2019. Architecture & Secure Delivery demand is higher in the later phase by IRR=3.17 (95% CI: 2.78–3.61, $p<.001$), and Operations demand is higher by IRR=1.93 (95% CI: 1.71–2.19, $p<.001$). This documents a significantly elevated level of employer-expressed demand in the post-2020 period; it does not constitute a formal test of slope change or acceleration in the rate of growth, which would require an interaction specification beyond the scope of a 15-observation annual series. The data are consistent with H4 as reframed.

5.2.5. H5: Phase reallocation

Because skill-family categories are not mutually exclusive at the posting level (a single posting may contribute to both "Cloud Security" and the broader "Security" baseline), the independence assumption required for valid chi-square inference does not hold. H5 is therefore assessed descriptively through the phase-share comparison in Table 9. The generic baseline's proportional share of total mentions contracted from 64.3% in 2010–2014 to 53.8% in 2020–2024, while specialized Architecture and Operations bundles steadily increased their proportional footprint across the three phases. This pattern is consistent with a gradual, evolutionary reallocation of employer-expressed demand rather than an abrupt structural disruption. The data are consistent with H5 as assessed descriptively.

To contextualize the large chi-square statistics, Table 9 presents the proportional share of total mentions for each skill family across the three phases. This reveals the substantive nature of the reallocation: the generic baseline's share of demand contracted from 64.3% to 53.8%, while specialized Architecture and Operations bundles steadily increased their proportional footprint over the 15-year window.

Table 9
Proportional share of skill family mentions by phase.

Skill Family	2010–2014 Share	2015–2019 Share	2020–2024 Share
Generic Baseline	64.3%	59.3%	53.8%
Governance	25.7%	25.0%	25.0%
Legacy/Offensive Controls	4.6%	6.5%	10.0%
Operations	4.3%	6.8%	7.9%
Architecture & Secure Delivery	1.1%	2.4%	3.3%
Total	**100.0%**	**100.0%**	**100.0%**

5.2.6. H6: Growth magnitude

Table 10 indicates substantial dispersion in growth rates across emerging competencies when growth is measured from each term's first meaningful appearance (≥100 mentions). The fastest-growing terms are DevSecOps (89.2%/year from 2016), zero trust (72.6%/year from 2020), and threat hunting (42.9%/year from 2016). The data are consistent with H6. These long-term growth metrics must be interpreted with caution regarding their base-rate sensitivity. Because emergent paradigms like DevSecOps and Zero Trust are calculated from near-zero or structural-zero base rates, their exceptionally high descriptive CAGRs (89.2% and 72.6%, respectively) reflect rapid early-stage market diffusion dynamics rather than sustained, large-scale structural dominance across the broader enterprise computing landscape.

5.2.7. H7: The data are consistent with H7 across four sensitivity checks

First, re-estimating family growth with negative binomial GLMs yielded directionally consistent results (see Table C1). Second, architecture and operations growth estimates remained stable in leave-one-year-out (jackknife) tests and exclusion periods (see Table C2). Third, a 'vocabulary tightening' check revealed that while the broad 'security' token is flat, the specific term 'cybersecurity' carries a strong growth

Table 10
Compound annual growth rates (CAGR).

Skill	Start year (≥100 mentions)	Start rate per 10k	2024 rate per 10k	CAGR to 2024
DevSecOps	2016	0.07	11.84	89.2%
Zero Trust	2020	0.09	0.81	72.6%
Threat Hunting	2016	0.14	2.51	42.9%
IAM	2013	0.05	0.77	28.3%
PAM	2011	0.13	2.42	25.0%
Cloud Security	2010	0.71	16.00	25.0%
Cybersecurity	2010	1.24	8.75	15.0%
NIST	2010	1.42	9.54	14.6%
ISO 27001	2010	1.15	6.48	13.2%
Threat Intelligence	2010	2.02	10.67	12.6%
Incident Response	2010	7.56	21.27	7.7%
SOC	2016	0.08	0.12	5.3%
Malware Analysis	2010	0.86	1.74	5.2%
Vulnerability Management	2010	6.85	13.53	5.0%
SIEM	2010	0.31	0.49	3.2%
Penetration Testing	2010	4.94	7.31	2.8%
Security Architecture	2010	5.78	7.78	2.1%
Risk Management	2010	149.86	193.42	1.8%
Encryption	2010	18.11	17.07	-0.4%
Security	2010	450.44	424.43	-0.4%
Ethical Hacking	2010	0.91	0.71	-1.8%
Digital Forensics	2010	7.30	5.46	-2.1%
Firewall	2010	53.86	31.66	-3.7%
Data Loss Prevention	2010	4.66	2.52	-4.3%

signal (IRR = 1.161, $p < .001$). Finally, the architecture-operations coupling (r=0.936) persisted even when SOC/SIEM labels were excluded, confirming the relationship is not driven by those specific umbrella tokens (see Table C3).

6. Discussion

The longitudinal pattern points to a shift in how employers articulate cybersecurity needs. General "security" language persists but no longer distinguishes specialized roles; postings increasingly couple architectural literacy, operational alignment, and formal compliance expectations. This is consistent with employers moving away from treating design, operations, and governance as separable silos toward integrated capability bundles, aligning with the human-capital, search-and-matching, and diffusion mechanisms outlined in Section 3.

One consequence is that the locus of scarcity migrates. Early in the period, scarcity is largely generic: organizations seek “security-minded” technologists. By the modern era, scarcity is specific to combinational skill sets that bind architecture, operations, and governance. A security architect fluent in cloud provider controls and identity orchestration, a DevSecOps/Cloud practitioner capable of embedding automated security guardrails and telemetry into a delivery pipeline, and a compliance lead who translates frameworks into control baselines and evidence routines represent complementary forms of capital that together become necessary. The data show these complements rising together rather than substituting for one another, supporting the view that secure operations at scale require bundles of capabilities rather than isolated expertise.

It is worth noting, however, that these mechanisms are not mutually exclusive. The same posting-language trend, for instance, rising DevSecOps mentions, is plausibly compatible with technology diffusion, evolving HR terminology conventions, compliance-mandate pressure, and broader software-engineering labor market shifts acting in combination. The present analysis documents consistency between the observed trends and each theoretical mechanism; it does not isolate their individual contributions, and readers should interpret the theoretical framing as a set of complementary lenses rather than competing hypotheses that the data resolve.

A critical finding of this longitudinal analysis is the divergence within operational competencies in employer hiring language, consistent with what practitioners describe as 're-platforming' of security work. Specifically, the data show that the SIEM label, once a cornerstone of security operations job descriptions, experienced a 27% decline in momentum after 2020. Similarly, the absolute prevalence of the 'SOC' umbrella term remained nearly invisible at 0.1 per 10,000 postings.

In contrast, the explosive growth of DevSecOps (89.2% CAGR) and Cloud Security (25.0% CAGR) in hiring language suggests that employers are increasingly specifying security as integrated directly into the software delivery pipeline and cloud-native architectures, rather than as standalone monitoring functions. This posting language shift is consistent with a transition in expressed employer priorities from reactive, manual log-analysis roles toward proactive, pipeline-integrated delivery competencies. However, the data reflect what employers are asking for in job advertisements, not direct evidence of implemented operational changes.

Consequently, the observed co-movement between architecture & secure delivery (r=0.934) is likely driven by pairing cloud-native design with automated response capabilities, rather than by legacy perimeter-and-SOC models. This linkage is further reinforced by the inclusion of DevSecOps within the architecture and delivery family, highlighting that modern 'secure-by-design' mandates require the tight integration of automated delivery pipelines with architectural controls. This suggests that for modern firms, 'operations' is no longer a separate function that occurs after deployment but rather a continuous architectural requirement, as evidenced by the high demand for integration-heavy skills such as threat hunting (42.9% CAGR) and identity-first controls.

When evaluating these longitudinal trajectories, we must consider alternative macro-level explanations beyond pure organizational maturation. First, changes in job posting conventions and HR technology have systematically institutionalized standardized corporate language. The widespread adoption of Applicant Tracking Systems (ATS) and programmatic job board cross-posting often inflates boilerplate text, such as compliance standards or generic certification requirements. This dynamic likely contributes to the lower relative volatility observed in the Governance family (CV = 0.057). Second, vendor taxonomy transformations in online job listing repositories can mimic shifts in organic demand. By conducting all 24 extractions in a single consistent session using the same standardized Lightcast skill taxonomy interface (documented in Appendix A3), we minimized cross-vintage taxonomy drift as a confound. Third, shifts in compliance pressures, including federal secure software mandates and stringent supply-chain certification requirements, have imposed non-discretionary pressure on employers. Finally, the broader trajectory of fluctuations in generic IT labor demand must be acknowledged. The post-2020 acceleration in cloud architecture hiring language is deeply intertwined with broader macro-level shifts in enterprise computing demand across the economy, meaning that expressed demand for cybersecurity skills is both a driver and a consequence of wider software engineering labor market trends.

6.1. Practical implications

Two practical implications follow. First, when budgets are constrained, the data argue for investing in identity, logging/telemetry, and incident response together, given their multiplicative effects on resilience and auditability. Second, program readiness should be jointly assessed across design, operations, and governance, for example, whether a student team can implement identity-aware cloud access, produce usable telemetry, and assemble control-aligned evidence. Table 11 summarizes these steps for academic, employer, and policy stakeholders.

This co-movement also has implications for curriculum design. Cloud security and identity concepts should be taught alongside incident response and SOC analytics so that students experience design and operations as a coherent system rather than separate units (Towhidi & Pridmore, 2023). Frameworks such as NIST CSF and ISO/IEC 27001 are best embedded as scaffolding for design decisions and operational evidence, rather than taught as standalone compliance topics.

Beyond localized curriculum and institutional design, these combinational findings provide empirical support for accelerating compliance and certification frameworks, most notably the Department of Defense’s Cybersecurity Maturity Model Certification (CMMC) program. By documenting a long-term shift in employer-expressed demand from isolated tool-specific language to institutionalized capability bundles, this study offers an empirically grounded reference point for CMMC stakeholder strategies. For example, CMMC Level 1 (“Foundational”) practices map

Table 11
Stakeholder implications.

Stakeholder	Implication Drawn from Results	Concrete Action
Academic Programs	Integration of architecture, operations, and governance is the market default.	Pair cloud/identity design with SOC/IR labs; embed NIST/ISO evidence artifacts in projects
Employers	Scarcity is in combinational skill bundles.	Write postings focused on outcomes and interfaces; fund cross-training and rotations among architecture, SOC, and GRC.
Policymakers & Consortia	Guidance that tightens integration outperforms a perimeter-only focus	Emphasize identity hygiene, logging depth, response rehearsal, and control assurance in programs and incentives

directly to our Legacy Controls family, which the data show are plateauing or contracting as baseline operational expenses. In contrast, CMMC Level 2 ("Advanced") requirements, which mandate institutionalized documentation, continuous vulnerability monitoring, and robust identity protections, align cleanly with the tightly coupled co-movement identified between our Operations and Governance skill bundles ($r = 0.718$, $p < .001$). Organizations aiming for CMMC alignment should abandon perimeter-focused hiring and instead design job roles that explicitly link access policy (PAM/IAM) to pipeline-integrated telemetry (DevSecOps), thereby directly facilitating the continuous evidence collection required by formal maturity assessments.

Another implication concerns timing and the cost of delay. The late, rapid rise of cloud security and zero trust suggests that organizations that deferred adoption now face steeper learning curves; their postings must specify multiple competencies simultaneously, intensifying talent competition. Conversely, organizations that internalized governance earlier, risk management, and standards alignment appear to experience smoother capability growth, consistent with institutionalization, which lowers variance and eases future adoption.

Threats to inference arise from the nature of the posting data and from platform dynamics. Postings reflect stated demand rather than realized task content; they can over-index signaling (e.g., certification acronyms) and understate tacit skill. Vendor taxonomies evolve, and measurement depends on year-aligned labels and transparent canonicalization. However, several features of the present design mitigate these threats: within-platform comparisons, fixed reporting parameters across years, explicit mapping rules, and results reported as shares per postings denominator when the denominator is available. The phase structure (i.e., generalist to specialist; architecture & secure delivery moving in tandem; governance normalization) persists under sensitivity analyses that remove the broad "security" token or tighten context rules for ambiguous terms, lending credibility to the core interpretation.

The results also inform debates about automation and AI in security operations. If telemetry volume and analytic tooling increase faster than hiring can, one might expect employer language to pivot away from human-centered operations. Instead, postings emphasize human-in-the-loop competencies such as incident handling, correlation logic, investigation, and evidence management, suggesting that analytic automation is complementary to, not a substitute for, practitioner skill at current maturity levels. This complements the human capital perspective: as tools improve, the returns shift toward workers who can orchestrate, interpret, and continuously improve those tools within governed architectures.

6.2. Limitations and scope conditions

Three limitations should temper the interpretation of these results. First, keyword matching is prone to contextual errors, particularly with broad tokens such as 'security' or ambiguous shorthand such as 'IR'. While mitigated through normalization and canonicalization, residual misclassification remains possible. Second, the dataset reflects a single vendor's pipeline, which may over-index certain subsectors or job levels. Third, temporal factors, such as economic cycles and the COVID-19 pandemic, affect both the volume and the phrasing of advertisements.

Crucially, readers must interpret these time-series findings as explicit labor-market demand signals rather than as direct, on-the-job evidence of implemented organizational practices or internal security maturity. Job advertisements document what employers look for at the point of vacancy; they do not measure actual workforce capability or operational security posture. Furthermore, while our 15-year longitudinal analysis successfully uncovers macro-level demand shifts, the underlying time series contains 15 discrete annual observations. This structural boundary requires appropriate caution to avoid overinterpreting statistical significance or model interaction effects. While the models demonstrate strong directional trends, the precise parameter metrics should be treated as informative approximations rather than absolute structural invariants.

Furthermore, the national roll-up intentionally averages over regional and sectoral variation to identify structural trends. Additionally, this study's controlled vocabulary was finalized before the rapid emergence of AI-specific security competencies (e.g., AI/ML security engineering, adversarial machine learning, LLM red-teaming) in the 2022–2024 hiring market. These boundaries suggest several productive avenues for further inquiry, which are detailed in Section 6.3 as a research agenda.

Finally, we acknowledge that our generic baseline token ('security') subsumes specialized compound phrases (e.g., 'cloud security' or 'information security') within the same posting. Because proprietary platform constraints limit the extraction of joint-probability co-occurrence matrices at the microdata level across 15 years, we could not calculate strictly non-overlapping proportional shares. Consequently, the relative contraction of the generic baseline reported in Table 9 may slightly understate the magnitude of the shift, as specialized terms mechanically inflate the generic denominator. Future research that uses raw microdata access should isolate these nested terms to quantify this reallocation further.

6.3. Directions for future research

The findings and methodological framework established in this study provide a scaffold for several critical avenues of inquiry. First, as noted in the limitations, the rapid emergence of AI-specific security competencies, such as adversarial machine learning, LLM red-teaming, and AI/ML security engineering, occurred after the current vocabulary was finalized. Future research should extend this 15-year series through 2027, using an expanded vocabulary, to evaluate whether these nascent AI signals follow a diffusion trajectory akin to the "early majority" phase observed in DevSecOps and cloud security.

Second, while this study identifies a "shift-left" in labor demand toward integrated delivery pipelines, it remains an open question whether this transition translates to measurable reductions in organizational risk. Future work should investigate whether these integrated capability bundles improve resilience or merely shift technical debt within the software development lifecycle.

Third, the regional and sectoral variation of these trends warrants deeper investigation. Disaggregating national data by industry (e.g., healthcare vs. finance) and metropolitan area could reveal where adoption is leading or lagging, providing more localized guidance for workforce development.

Finally, as AI-driven automation increasingly permeates Security Operations Centers (SOC), longitudinal research must track the "longevity" of human-in-the-loop skill signals. Scholars should monitor whether competencies such as investigation and evidence management remain resilient complements to automation, or undergo a new wave of obsolescence as autonomous agents mature.

7. Conclusion

This study provides a 15-year empirical record of how employer-expressed demand language for cybersecurity competencies has changed. The data show a qualitative shift toward integrated capability bundles, with DevSecOps and Cloud Security now appearing as baseline recruitment expectations in employer hiring language, consistent with what diffusion theory would describe as an early-majority phase of adoption.

Theoretically, this research advances Human Capital Theory and Diffusion of Innovations within the cybersecurity domain by quantifying the co-movement between expressed architectural preferences and operational resilience requirements in job advertisements. We contribute to the body of knowledge by demonstrating that within corporate recruitment signals, the desired profile has shifted from generalist "security-mindedness" to a stated preference for "integration-

fluency," the documented market demand for practitioners who can conceptually cross-navigate automated telemetry, identity-first controls, and standards-based governance.

For researchers, this establishes a longitudinal baseline for studying the durability of these skill signals and their link to risk reduction (Section 6.3). For practitioners, it offers a starting point for aligning hiring and curricula with secure-by-design expectations.

Declaration of Generative AI and AI-Assisted Technologies in the Writing Process

Conceptual design and visual layout for Fig. 1 were prototyped using Google Gemini. The final graphic was manually recreated and refined by the author in Microsoft PowerPoint to ensure technical accuracy and stylistic alignment. The author remains fully responsible for the final content and interpretation.

The Prompt

"Create a high-quality academic schematic diagram for a cybersecurity research paper titled 'The Evolution of Cybersecurity Skill Demands.' The diagram should show two phases. On the left, labeled 'Early 2010s: Foundational Era,' show three separate, siloed rectangular boxes: 'Generic Security,' 'Firewalls,' and 'Encryption.' On the right, labeled '2020s: Security-by-Design,' show a Venn diagram where three circles overlap. Label the circles: 'Architecture & Secure Delivery (Cloud, DevSecOps, Zero Trust),' 'Operations (Threat Hunting, IR),' and 'Governance (NIST, ISO 27001).' Between the two phases, place a large arrow pointing right, labeled 'Structural Maturation and Re-platforming.' Use a clean, professional, and minimalist academic style with a white background and grayscale or blue-tinted professional colors."

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CRediT authorship contribution statement

Joseph H. Schuessler: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Joseph H. Schuessler reports administrative support was provided by Tarleton State University. Joseph H. Schuessler reports a relationship with Tarleton State University that includes: employment. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.cose.2026.105212.

Appendix A. Complete controlled vocabulary

This appendix details the definitional and chronological parameters for the 24 cybersecurity competencies tracked in this study. Table A1 presents the controlled vocabulary, illustrating how representative job posting variants and acronyms were mapped to canonical labels to ensure construct validity. Table A2 documents the emergence timeline and taxonomy status for each concept, distinguishing between foundational terms that are tracked continuously from the 2010 baseline and emergent terms treated as structural zeros until their first organic appearance.

Table A1
Controlled vocabulary - canonical concepts and representative posting variants.

Canonical Label	Representative Variants and Acronyms Mapped to Label
Architecture & Secure Delivery	
Cloud Security	cloud security, cloud-security, security in cloud, cloud access security
Zero Trust	zero trust, zero-trust, ZTNA
DevSecOps	DevSecOps, dev-sec-ops, security in DevOps
Security Architecture	security architecture, infrastructure security, network architecture *(included as cybersecurity-specific network design context, not generic IT networking)*
Operations	
Incident Response	incident response, IR, incident handling
SIEM	SIEM, security information and event management
SOC	SOC, security operations center, SOC analyst
Threat Intelligence	threat intelligence, threat intel, CTI
Threat Hunting	threat hunting, cyber threat hunting
Vulnerability Mgmt	vulnerability management, vulnerability assessment, VMS
Malware Analysis	malware analysis, reverse engineering malware
Digital Forensics	digital forensics, computer forensics, forensics
Governance	
Risk Management	risk management, enterprise risk, risk assessment *(risk assessment is a core sub-practice of risk management per NIST CSF and ISO 27001 frameworks)*
NIST	NIST, NIST CSF, NIST 800-53, NIST 800-171
ISO 27001	ISO/IEC 27001, ISO-27001
IAM	identity and access management, identity management, IDM
PAM	Privileged access management, privileged account management, PIM
Data Loss Prevention	data loss prevention, DLP
Legacy/Offense	
Penetration Testing	penetration testing, pen test, pentesting
Ethical Hacking	ethical hacking, CEH
Firewall	firewall, network security *(included only in posting where network security co-appears with perimeter/boundary protection context; see* Section 4.6)
Encryption	encryption, cryptography

Note: Variants listed are representative examples of how each canonical concept appears in Lightcast job posting text. Lightcast's skill taxonomy applies variant-matching internally at the platform level; this table illustrates the conceptual scope of each canonical label rather than enumerating every string matched.

Table A2
Concept emergence and taxonomy status. (Proxy based on First Organic Appearance ≥100 Mentions).

Canonical Label	First Organic Appearance	Treatment in Analysis	Measurement Basis & Taxonomy Alignment
Generic Baseline (Security, Cybersecurity)	Pre-2010	Continuous	Standard Lightcast skill lookup
Architecture Baseline (Cloud Security, Sec. Architecture)	Pre-2010	Continuous	Standard Lightcast skill lookup
DevSecOps	2016	Pre-2016 treated as structural zero	Standard Lightcast skill lookup
Zero Trust	2020	Pre-2020 treated as structural zero	Standard Lightcast skill lookup
Operations Baseline (IR, Malware, Vuln Mgmt, Threat Intel)	Pre-2010	Continuous	Standard Lightcast skill lookup
SOC	2016	Pre-2016 treated as structural zero	Standard Lightcast skill lookup
Threat Hunting	2016	Pre-2016 treated as structural zero	Standard Lightcast skill lookup
Governance Baseline (Risk Mgmt, NIST, ISO 27001)	Pre-2010	Continuous	Standard Lightcast skill lookup
PAM	2011	Pre-2011 treated as structural zero	Standard Lightcast skill lookup
IAM	2013	Pre-2013 treated as structural zero	Standard Lightcast skill lookup
Legacy/Offense Baseline (Firewall, Encryption, Pen Testing)	Pre-2010	Continuous	Standard Lightcast skill lookup

Note on Duplicate and Context Management: De-duplication of unique postings is performed upstream by Lightcast's own platform pipeline (Lightcast, 2023a, 2023b); this study relies on that vendor-level de-duplication rather than applying a separate algorithm of its own.

Appendix A3. Extraction method and data provenance

The following describes the extraction method applied to each of the 24 canonical concepts. All 24 were extracted via (a) Lightcast's standardized skill taxonomy interface, applied uniformly in a single session (October 24, 2025). All 24 then underwent (c) post-export canonical merging; case-folding, Unicode normalization, hyphenation standardization, and assignment of variant strings to canonical labels per Table A1. Method (b), raw keyword matching against posting microtext, was not used; Lightcast's taxonomy handles variant-matching internally. Method (d), within-posting co-occurrence filtering, was not applied because aggregated Lightcast skill-count exports do not provide access to posting-level microtext; construct validity was managed instead through vocabulary-definition decisions, as described in Section 4.6. The complete annual extraction results (mention counts, total-postings denominators, and normalized rates) for all 24 canonical concepts are provided in full in Supplementary Data S1.

Appendix B. Annual posting totals (Denominator Data)

Year	Total Unique Postings (N)	Data Extraction Notes (all years: Lightcast JPA, all occupations, no skill filter, Newly Posted, national scope, extracted January 15, 2025; annual value = Lightcast "Unique Postings" figure, which represents the sum of monthly de-duplicated unique posting counts for the calendar year)
2010	12,465,406	See column header note.
2011	15,266,281	See column header note.
2012	15,076,470	See column header note.
2013	20,808,474	See column header note.
2014	22,386,635	See column header note.
2015	24,936,264	See column header note.
2016	26,390,568	See column header note.
2017	26,251,967	See column header note.
2018	33,195,573	See column header note.
2019	34,490,161	See column header note.
2020	34,733,422	See column header note.
2021	43,719,243	See column header note.
2022	48,599,774	See column header note.
2023	38,904,235	See column header note.
2024	36,892,559	See column header note.
Total (2010-2024)	434,117,032	Sum across years

Appendix C. Robustness and sensitivity analyses

This appendix reports robustness checks designed to evaluate whether the study's core findings depend on (i) estimator choice, (ii) temporal influence from any single year or macro shock period, or (iii) reliance on broad or potentially ambiguous tokens. Because the analysis is based on Lightcast's aggregated skill-mention counts (not posting-level microtext), robustness is evaluated using alternative vocabularies and operational definitions feasible under aggregated data constraints. Table C1, Table C2, Table C3.

Appendix C1. Estimator robustness (Poisson vs. negative binomial)

Table C1 presents the primary Negative Binomial GLM estimates alongside Poisson GLM estimates for comparison. Dispersion parameters (α) are small but non-trivial for the Architecture and Operations families, confirming the appropriateness of the NegBin specification as primary. Substantive conclusions are consistent across both estimators.

Table C1
Estimator robustness: annual growth (IRR per year) by Skill Family, 2010–2024 (N=15 years).

Skill family	Poisson IRR	95% CI	p-value	NegBin IRR	95% CI	p-value	α (dispersion)
Architecture & Secure Delivery	1.124	[1.069, 1.125]	<.001	1.141	[1.097, 1.186]	<.001	0.0276
Operations	1.049	[1.027, 1.071]	<.001	1.067	[1.045, 1.089]	<.001	0.0277
Governance	1.014	[1.004, 1.023]	.004	1.015	[1.005, 1.024]	.003	0.0065
Legacy/Offensive Controls	0.984	[0.978, 1.000]	.045	0.992	[0.981, 1.003]	.140	0.0086
Generic Security language	0.998	[0.989, 1.006]	.587	0.999	[0.990, 1.009]	.863	0.0065

Note: IRR > 1 indicates positive annual growth in mention rates net of overall posting volume (offset). The negative binomial results are directionally consistent with Poisson estimates.

Appendix C2. Temporal robustness (leave-one-year-out; excluding 2020–2021)

Table C2 evaluates sensitivity to temporal influence using (i) leave-one-year-out estimation (jackknife) and (ii) re-estimation excluding 2020–2021. Results are stable: architecture & secure delivery growth estimates remain tightly bounded, and architecture–operations co-movement remains strong.

Table C2
Temporal robustness: IRR stability and correlation stability.

Panel A. Leave-one-year-out IRR ranges (Poisson GLM; N=14 each run)
(Poisson GLM used for jackknife runs for computational comparability; NegBin estimates (Panel B) confirm directional stability)

Skill family	IRR min	IRR max
Architecture & Secure Delivery	1.120	1.131
Operations	1.043	1.056
Governance	1.011	1.017
Legacy/Offensive Controls	0.981	0.989
Generic Security Language	0.994	1.000

Panel B. Excluding 2020–2021 (Poisson GLM; N=13 years

Metric	Estimate
Architecture & Secure Delivery IRR	1.122
Operations IRR	1.048
Pearson r (Architecture, Operations)	0.930
Spearman ρ (Architecture, Operations)	0.879

Panel C. Leave-one-year-out correlation ranges

Correlation metric	Min	Max
Pearson r (Architecture & Secure Delivery, Operations)	0.923	0.947
Spearman ρ (Architecture, Operations)	0.807	0.908

Note. Correlations are computed on annual family-level rates per 10,000 postings.

Appendix C3. Vocabulary tightening and operationalization sensitivity

This set of checks evaluates whether findings depend on broad or ambiguous tokens and whether the core co-movement result persists under alternative definitions aligned with the paper's theoretical claims.

Table C3
Vocabulary and family-definition sensitivity.

Robustness Check	What Changes	Key Result(s)	Interpretation
Vocabulary Tightening (Remove Broad "Security" Token)	Generic baseline redefined from security + cybersecurity to cybersecurity-only	Generic baseline becomes strongly positive: IRR = 1.161, 95% CI [1.126, 1.197], p < .001	"Security" functions as a high-frequency background term; "cybersecurity" carries the growth signal. Main architecture & secure delivery conclusions remain intact.
Operations Definition Excluding SOC/SIEM Labels	Operations family excludes SOC and SIEM umbrella labels; retains IR/threat/vuln/hunting	Operations_alt IRR ≈ 1.049 (very similar to primary); Pearson r (Architecture, Operations_alt) = 0.936	The architecture & secure delivery coupling persists even when SOC/SIEM labels are excluded, indicating the relationship is not driven by those umbrella tokens.
Governance Definition Excluding Potentially Broad "Risk Management"	Governance family excludes "risk management" and uses NIST/ISO/IAM/PAM/DLP	Governance_alt IRR ≈ 1.071	Governance growth is partly masked by a broad, more stable "risk management" term; narrower governance signals increase more sharply.
Sensitivity: IAM/PAM reclassified as Architecture	IAM and PAM are moved from the Governance family into the Architecture family. This expands the Architecture group and narrows the Governance group to pure risk/compliance (Risk Mgmt, NIST, ISO).	Expanded Architecture: IRR = 1.167, 95% CI [1.141, 1.194], $p < .001$ Reduced Governance: IRR = 1.015, 95% CI [1.006, 1.024], $p = .001$	The core finding strengthens: modernizing delivery pipelines (Arch) accelerates even faster when identity frameworks (IAM/PAM) are bundled within it. The pure governance baseline remains stable and slow-growing, isolated from the explosive growth of identity engineering.

Note: These tests operationalize the intended "context tightening" principle at the level feasible for aggregated Lightcast outputs, by tightening vocabulary and family definitions rather than applying within-posting token windows.

Data availability

Aggregated annual counts and rates are provided in Supplementary Data S1. Underlying Lightcast posting-level microdata are proprietary and cannot be shared.

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